Why PostgreSQL (pgvector)

Pinecone

Weaviate

ChromaDB

Milvus

Vespa

Specialized vector DBs = Extra complexity

Hard to integrate with relational/metadata

Vendor lock-in and cost concerns

Handles structured + vector + text

Mature, reliable, ACID compliant

Open-source, cost-effective

Runs everywhere (local, cloud, containers)

Unified Queries

```
SELECT *
FROM documents
WHERE metadata->>'category' = ‘hr_policy’
ORDER BY embedding <#> query_vector
LIMIT 5;
```

SQL + Vectors + Metadata

Vector Indexes

HNSW IVFFlat

Deployment Advantage

Docker + pgvector setup

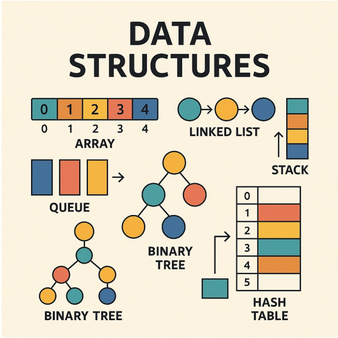
Storage Health Initialization

Vector Indexes

[-0.1848554 -0.17605719
0.18991971 0.4797751
0.04560479 0.00489019
0.0982461 -0.07267588]

Vectorization

Embedding
Model



Retrieval

Relevant Data

Build

Fast

Efficient

PGVector

IVFFlat Inverted File Flat

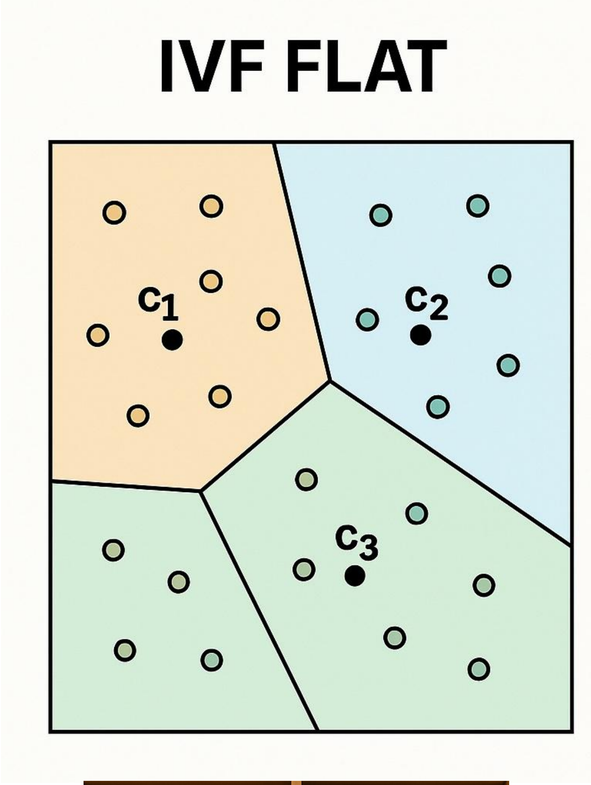
HNSW Hierarchical Navigable Small World

Memory-Constrained

Fast Builds

Trade Accuracy

Temp or Experimental

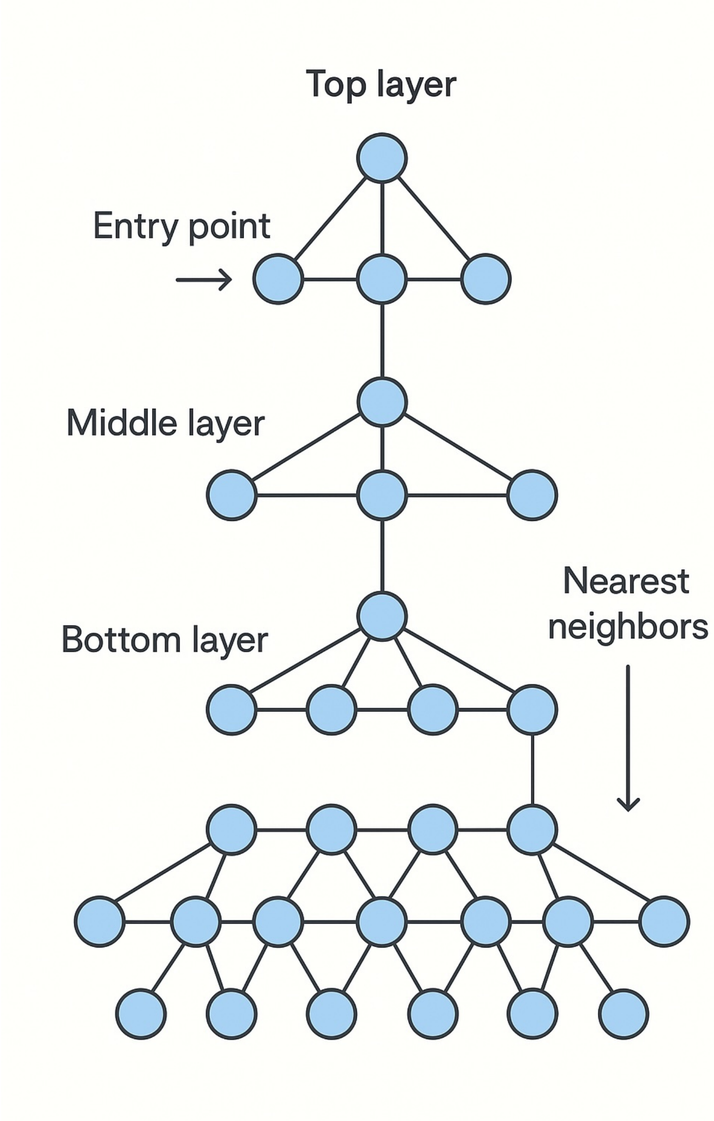


HNSW

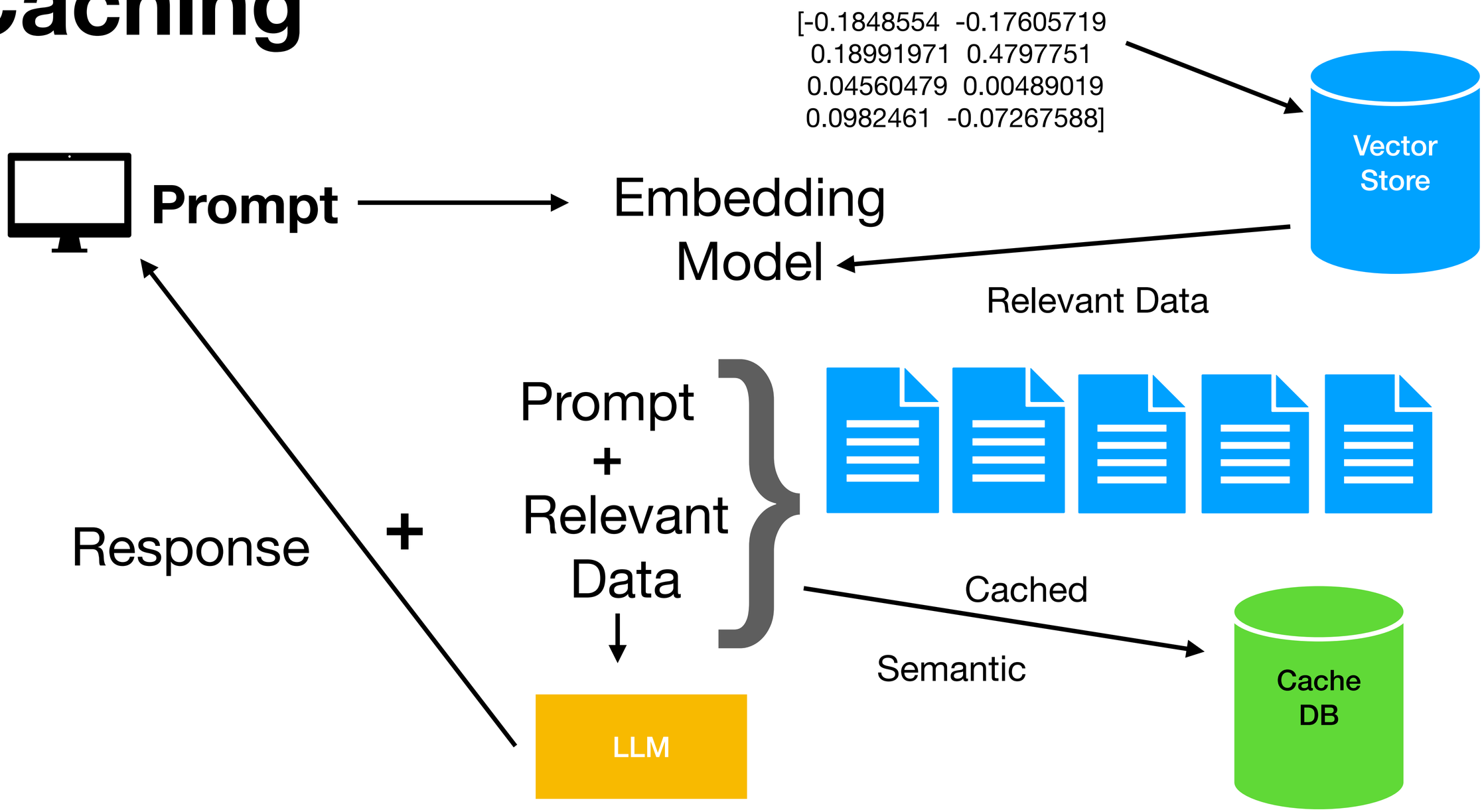
Best Recall

Fast Queries

Production Scale



Caching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0.04560479 0.00489019
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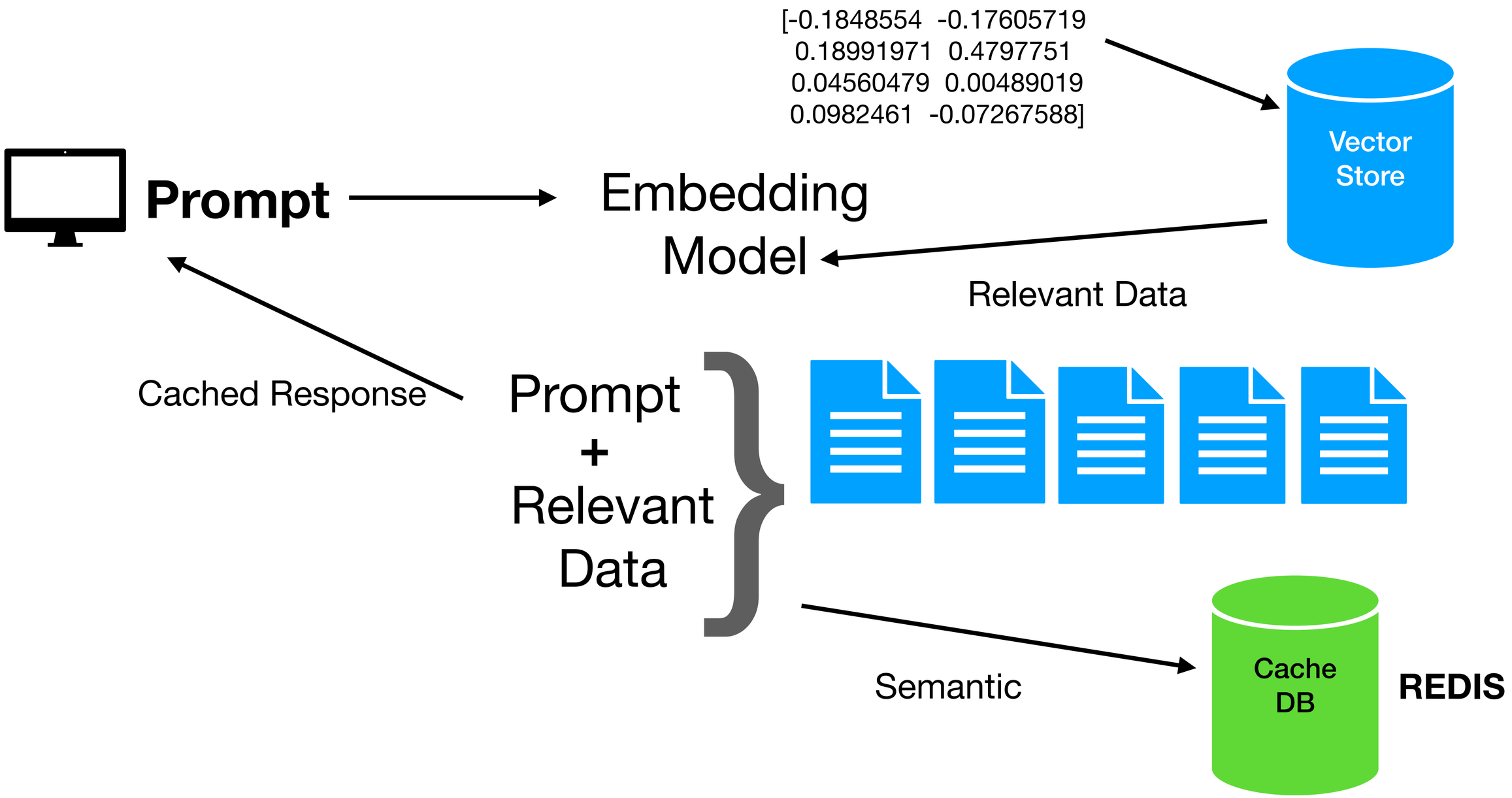
Vector Store
Prompt
Embedding Model
Relevant Data
Prompt + Relevant Data
Response
+
Cached
Semantic
Cache DB
LLM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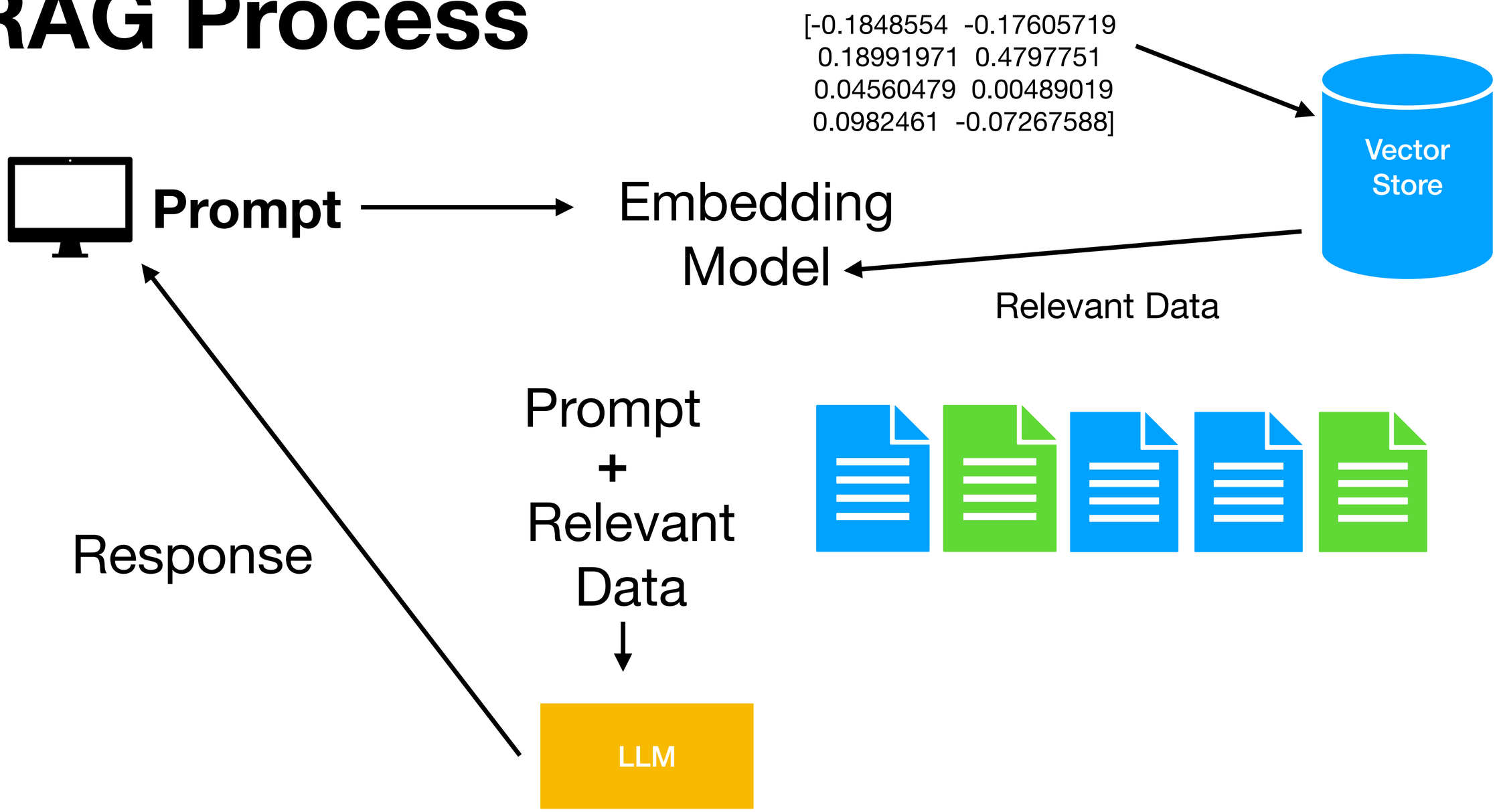
Vector Store
Prompt
Embedding Model
Relevant Data
Cached Response
Prompt + Relevant Data
Semantic
Cache DB
REDIS

RAG Process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Vector Store
Prompt
Embedding Model
Relevant Data
Prompt + Relevant Data
Response
LLM

RE-Ranking

Types

Simple Similarity Scores

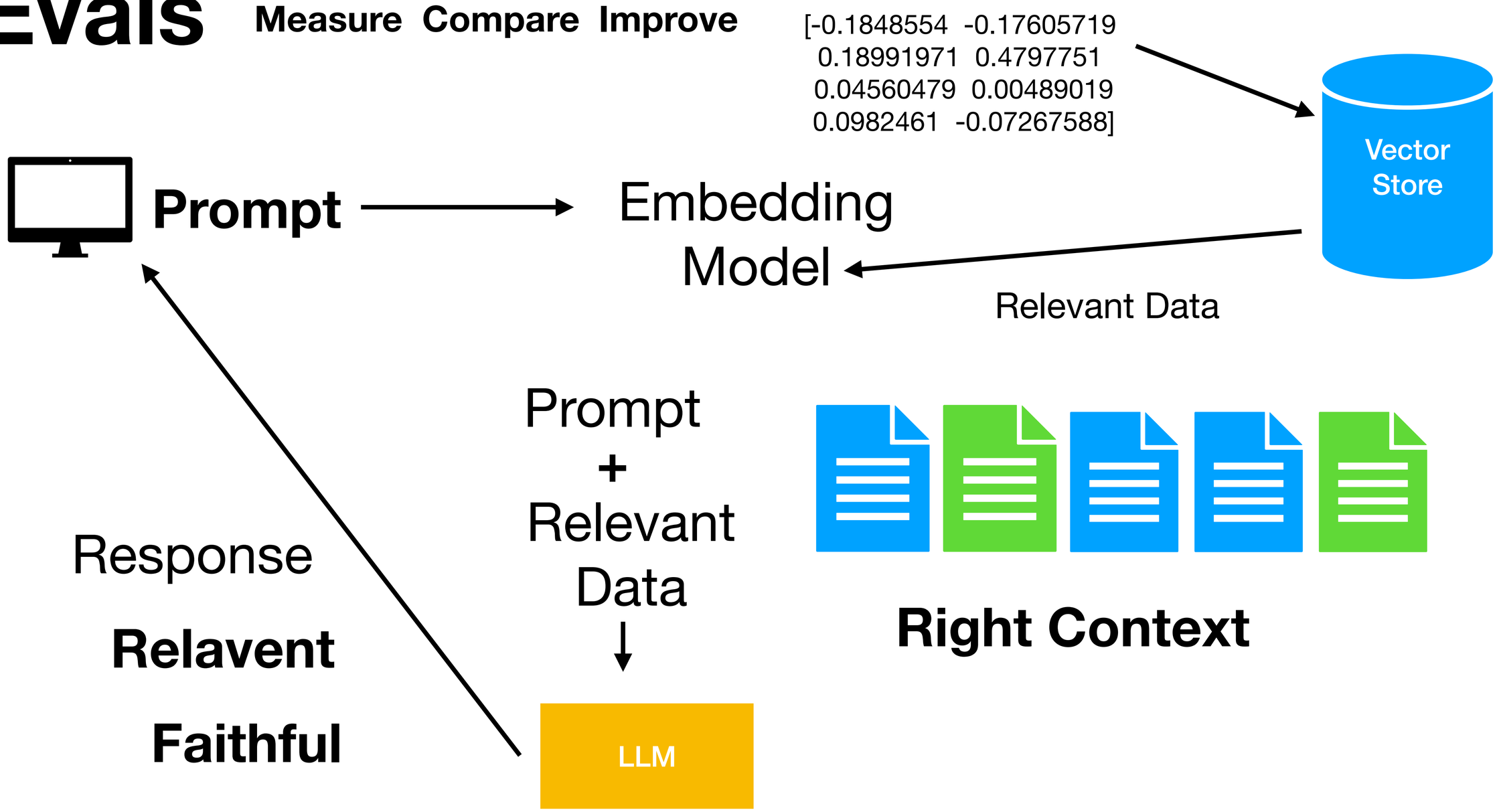
Cross-Encoder Reranking

Cohere Rerank

LLM-based Reranking

Evals

Measure Compare Improve



RAGAS

Question

Answer

Contexts

Reference Answer

Metrics

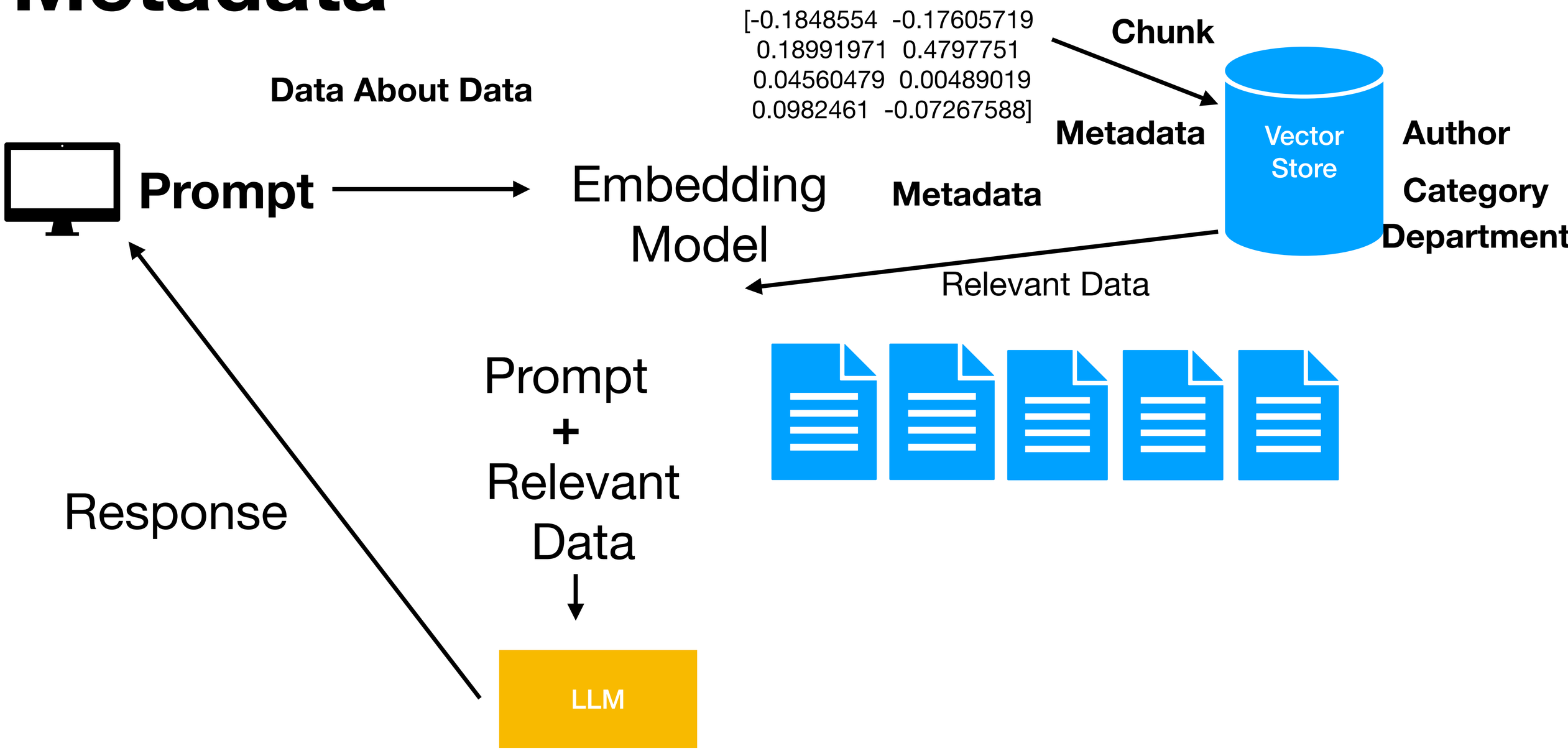
Faithfulness

Answer Relevancy

Context Precision

Context Recall

Metadata


Data About Data
Prompt
Embedding Model
[-0.1848554 -0.17605719
0.18991971 0.4797751
0.04560479 0.00489019
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Chunk
Metadata
Metadata
Vector Store
Author
Category
Department
Relevant Data
Prompt
+
Relevant
Data
Response
LLM

Semantic Search

Precise Targeted Context Aware

Common Metadata

Source Information	File Name	Url
Categorical Tags	Department	Doc Type
Timestamps	Created_At	Last_Updated
Custom Fields	Category	Version